TUTORIAL

Checking and exploring umbrella sampling runs

> Draft, 2026-10-23. The figures were made with the development version of EasyHybrid 3.2.0.

An umbrella sampling (US) run leaves dozens or hundreds of windows on disk. WHAM turns them into a free energy profile (PMF), but WHAM does not tell you whether the windows were good: whether neighbouring windows overlap, whether each window stayed near its target, whether the run is long enough. The **Umbrella Sampling Browser** answers those questions and lets you load the structures of any window, a range of windows or a path on a 2D PMF as objects in EasyHybrid.

This tutorial uses a small system that runs in minutes: the gas-phase SN2 reaction

$Cl^- + CH_3Br \rightarrow ClCH_3 + Br^-$

with the AM1 Hamiltonian (6 atoms, charge −1). You will open three runs:

Run	Reaction coordinate	Windows	Force constant	What it shows
`SN2_US_1D`	d(C–Br) − d(C–Cl)	31	600 kJ mol^{-1} $Å^{-2}$	a well-behaved 1D run
`SN2_US_1D_stiff`	the same	31	4000 kJ mol^{-1} $Å^{-2}$	windows that do not overlap
`SN2_US_2D`	d(C–Br) and d(C–Cl)	15 × 15	600 kJ mol^{-1} $Å^{-2}$	a 2D run, its PMF and the reaction path

The three runs, the PES scans they start from and the starting structure are in `examples/tutorials/11_umbrella_browser_sn2/`, so you can follow the tutorial without running anything (section 2). Section 1 explains how they were made, if you want to run them yourself.

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1. (Optional) Running the example yourself

Each US window starts from one frame of a relaxed PES scan, so a US run in EasyHybrid always has two steps: a scan along the reaction coordinate, then the US from the scan frames.

The system. *File* → *Open* `examples/tutorials/11_umbrella_browser_sn2/sn2_cl_ch3br.xyz` (the reactant complex: Cl^- 3.2 Å from the carbon, on the side opposite to Br). Select all atoms, *Selection* → *Set as QC Atoms*, and in *System* → *Setup QC Region* choose the semiempirical method **AM1**, charge **−1**, multiplicity 1.

The scan. *Simulate* → *Reaction Coordinate Scans*. Reaction Coordinate 1: Coordinate Type **multiple distance**, Atom 1 = Br, Atom 2 = C, Atom 3 = Cl (pick them in the 3D view and press *Import from Picking Selection*), so that RC = d(C–Br) − d(C–Cl). Initial Distance −1.4 Å, Step Size 0.1 Å, Number of Steps 31, Force Constant 4000. The scan

takes a few seconds; the minimum of the reactant complex is at RC ≈ −0.9 Å and the top of the profile at RC ≈ +0.3 Å.

The umbrella sampling. *Simulate* → *Umbrella Sampling*, with the scan trajectory as Input Trajectory and the same reaction coordinate:

Field	1D run	2D run
Reaction Coordinate 1	d(C–Br) − d(C–Cl), as in the scan	simple distance Br–C, from 1.8 Å, 15 steps of 0.1 Å
Reaction Coordinate 2	—	simple distance C–Cl, from 1.7 Å, 15 steps of 0.1 Å
Force Constant	600 (and 4000 for the "stiff" run)	600
Integrator	Langevin, 300 K, collision frequency 25 ps^{-1}	the same
Time Step	0.001 ps	0.001 ps
Equilibration	2000 steps	1000 steps
Data Collection	10 000 steps; ptRes every step, ptGeo every 100	4000 steps; the same
Number of CPUs	12	12

With 12 processes the 1D runs take about 1.5 min each and the 2D run about 5 min (the 2D run starts from a 2D scan of the two distances over the same grid). Then run *Analysis* → *WHAM* on each run (100 bins for 1D, 60 × 60 for 2D, 300 K): its `*_pmf.log` file goes to the `wham` folder of the run, where the browser finds it.

Each run is a folder named after the trajectory, here `SN2_US_1D`, with this content:

Folder or file	Content
`output.log`	all the parameters of the run
`optmization.ptGeo/`	the starting structure of each window
`equilibration/`	the equilibration frames, one `window*.ptGeo` per window
`data_collection/`	the reaction coordinate at every step (`window*.ptRes`), used by WHAM
`data_collection_traj/`	the data collection frames (`window*.ptGeo`)
`wham/`	the WHAM output (`output_pmf.log`)

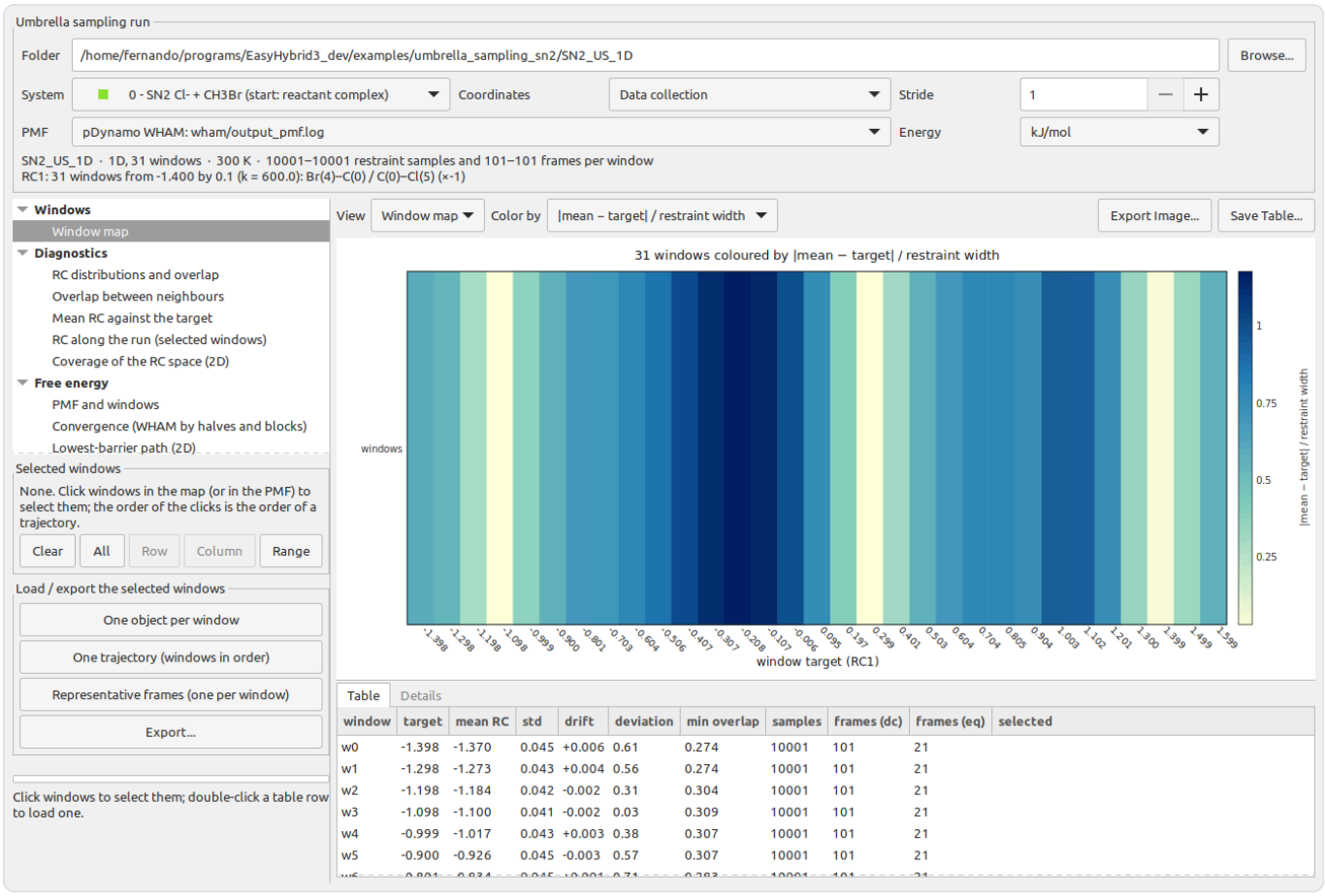
2. Opening a run

Load the system first (*File* → *Open* `sn2_cl_ch3br.xyz`): the browser loads frames as new objects of a system with the same atoms. Then open **Analysis → Umbrella Sampling Browser...** (or press **Browse Windows...** in the WHAM window, which opens the browser on the run chosen there).

Press *Browse...* and choose `examples/tutorials/11_umbrella_browser_sn2/SN2_US_1D`. The browser reads the folder and shows, under the controls, what it found: a 1D run with 31 windows at 300 K, 10 001 restraint values and 101 frames per window, and the definition of RC1 (`Br(4)-C(0) / C(0)-Cl(5) (×-1)` , the weighted distance d(C–Br) − d(C–Cl), with the 0-based atom indices of pDynamo).

Control	What it does
System	the system that receives the loaded frames (it must have the same atoms)
Coordinates	Data collection, Equilibration or the initial structure of each window
Stride	read every n-th frame of each window
PMF	a PMF file of pDynamo's WHAM, or a WHAM computed by the browser
Energy	kJ/mol or kcal/mol

The left side has the views (*Windows*, *Diagnostics*, *Free energy*), the list of selected windows and the buttons that load or export them. The right side shows the plot of the view, a table and a *Details* tab with the explanation of the numbers.



The window map of the 1D run, coloured by the deviation of each window from its target

3. Are the windows good?

The window map

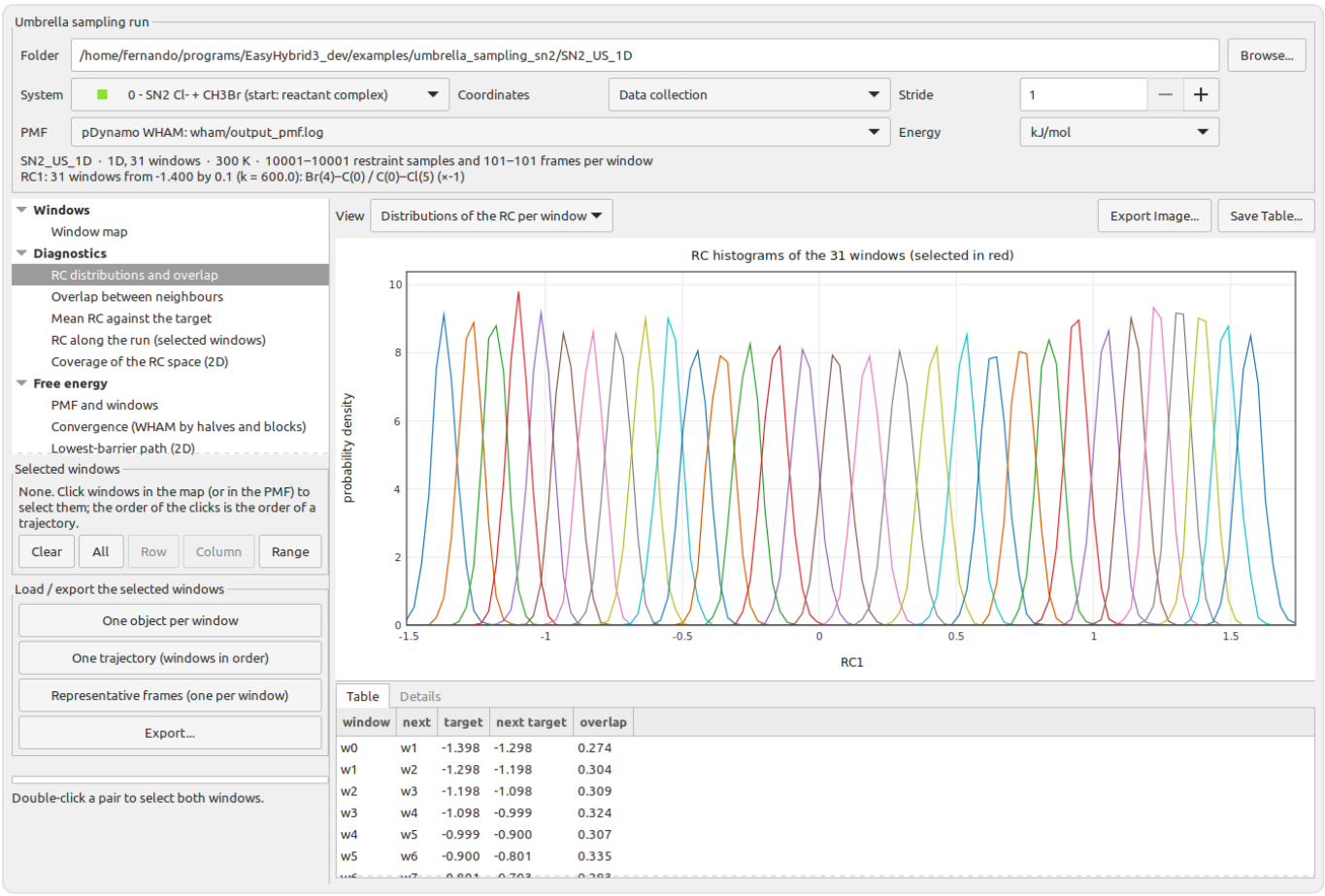
Windows → *Window map* shows one cell per window (a strip for 1D, a matrix for 2D), coloured by the quantity chosen in *Color by*:

Color by	Meaning	Warning sign
Frames	coordinates saved in the window	a window with fewer frames than the others (a failed window)
|mean − target| / restraint width	how far the window's mean RC is from its target, in units of the width sqrt(kT/2k) the restraint alone would give	above ~3: the window slid away from its target
Drift	change of the mean RC between the first and the second half	a large drift: not equilibrated
RC spread	standard deviation of the RC in the window	
Lowest overlap with a neighbour	see below	below 0.03
PMF at the target	the PMF where the window was centred	

The table under the plot has every number per window. In `SN2_US_1D` all windows stay within 1.2 widths of their targets.

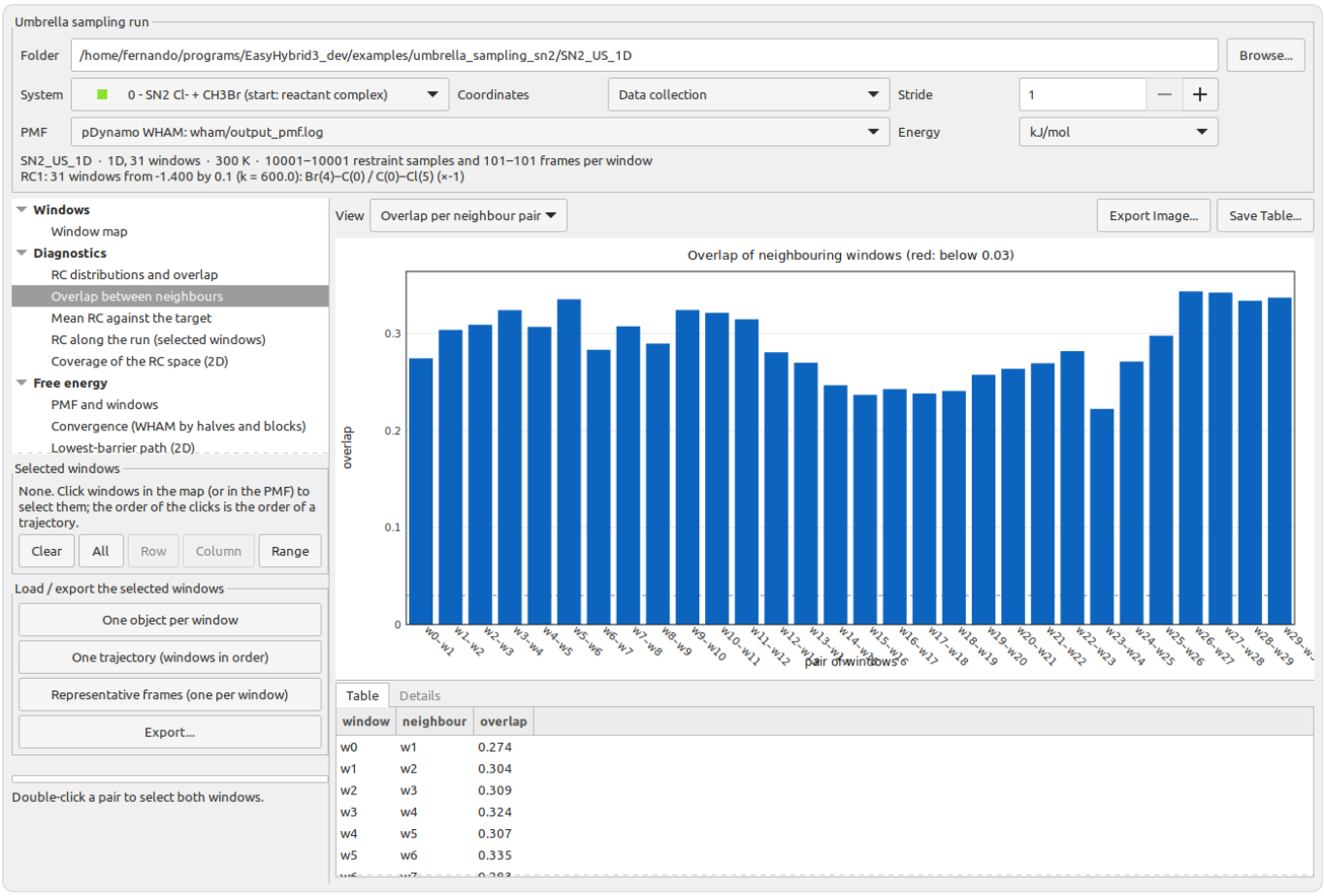
Overlap of neighbouring windows

WHAM can only connect two windows if they sample some common values of the reaction coordinate. *Diagnostics → RC distributions and overlap* shows the distribution of the RC in every window, and *Overlap between neighbours* the overlap coefficient of each pair of neighbours: the common area of their two normalized histograms, 1 for identical distributions and 0 for disjoint ones.



RC distributions of the 31 windows of the 1D run with k = 600: every pair of neighbours overlaps

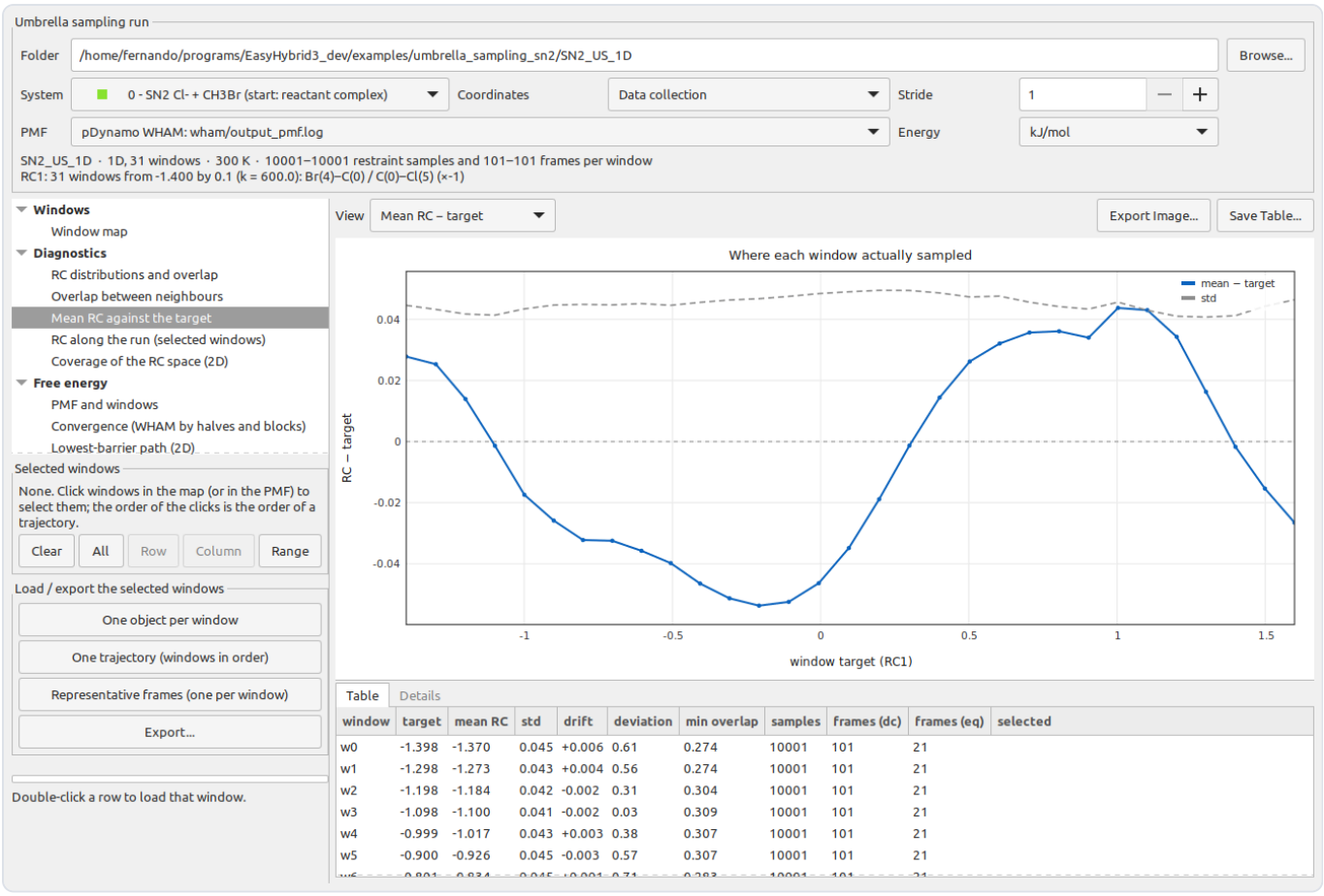
With k = 600 kJ mol^{-1} Å^{-2} each window spreads over about 0.045 Å (one standard deviation) and the targets are 0.1 Å apart: the distributions touch, and the lowest overlap of a pair is 0.22.



Overlap of each pair of neighbouring windows in the 1D run

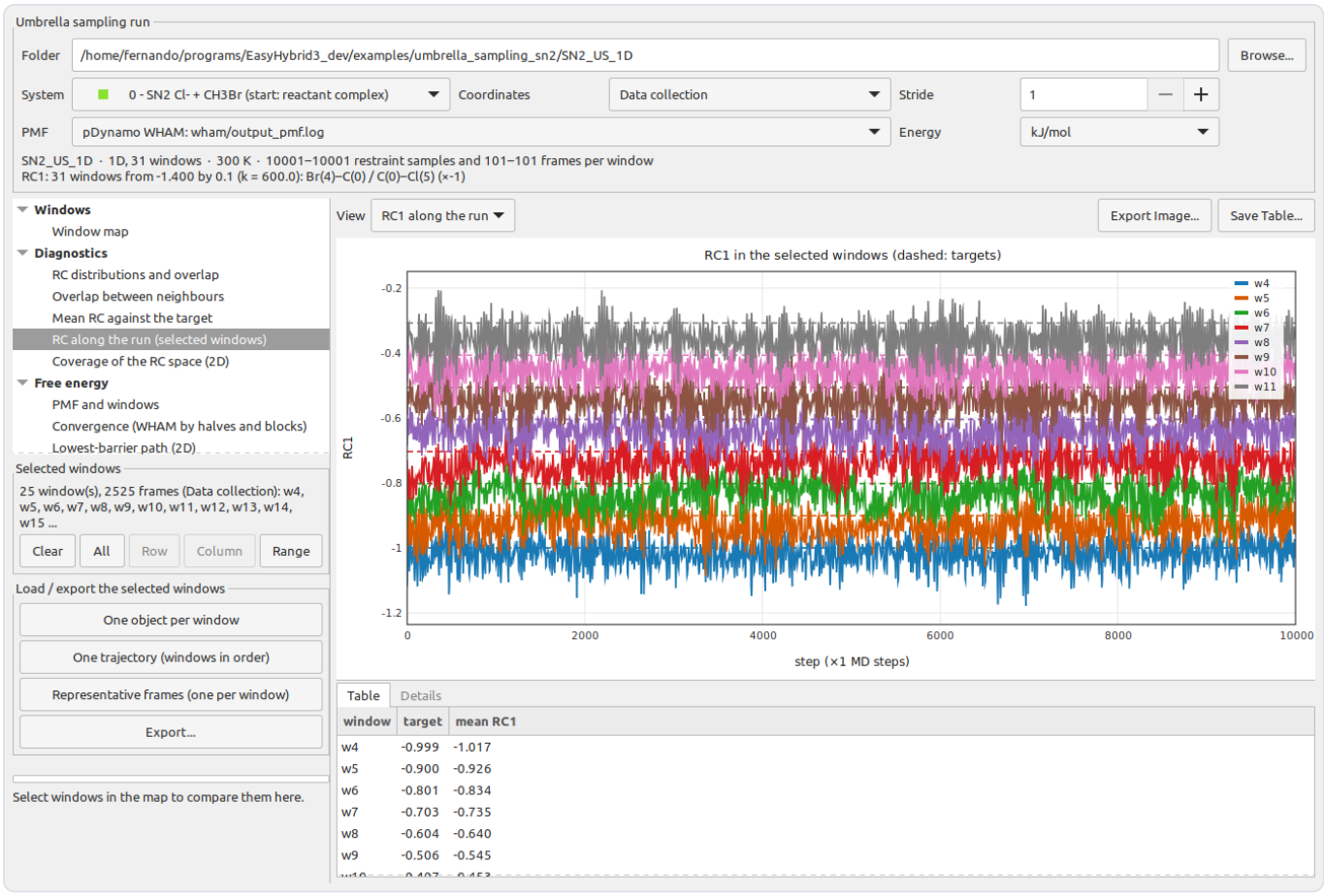
Mean RC against the target

Diagnostics → Mean RC against the target plots, for each window, where it actually sampled. The mean is pulled a few hundredths of an ångström downhill, towards the reactant and product complexes: the restraint does not cancel the slope of the PMF. That is normal; WHAM corrects for it. The *Drift* view shows the change of the mean between the two halves of the run.



Mean RC − target and the RC spread of each window

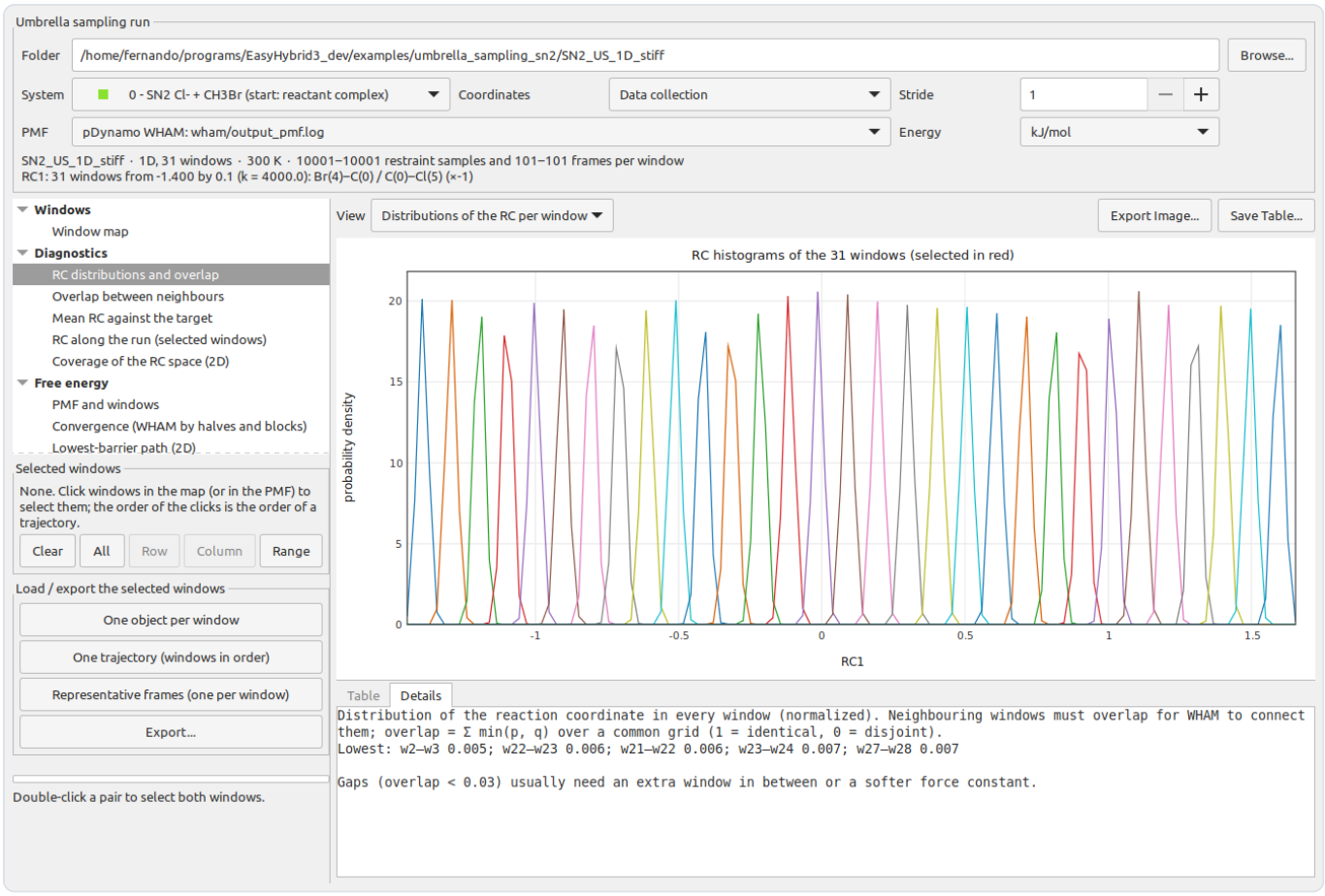
Diagnostics → RC along the run shows the RC at every step of the selected windows, with their targets as dashed lines (select windows first, section 5; with none selected it shows the middle window).



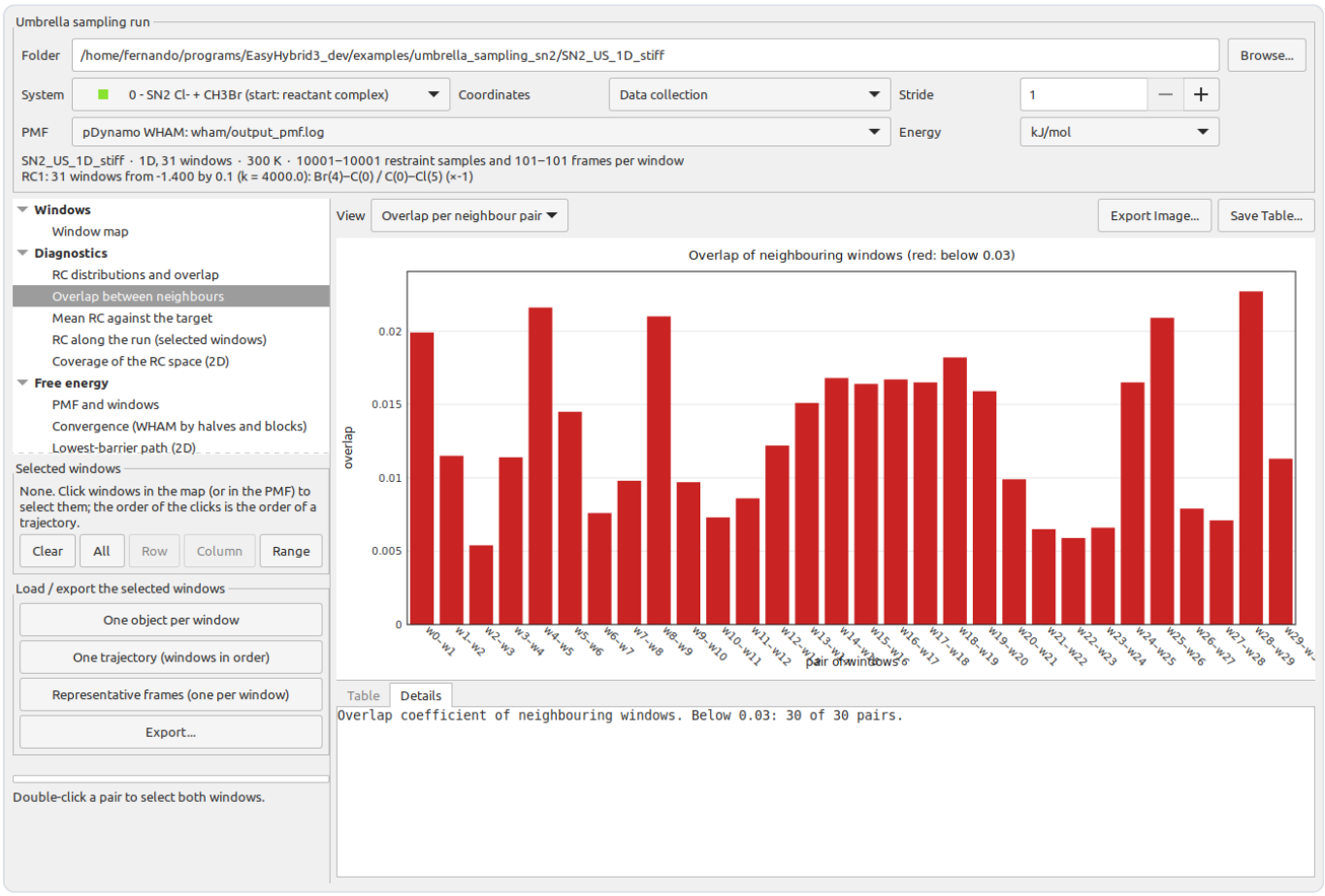
The reaction coordinate along the data collection of selected windows

A run whose windows do not overlap

Open `SN2_US_1D_stiff`, the same run with k = 4000 kJ mol^{-1} Å^{-2}. Each window now spreads over only 0.018 Å, five times less than the distance between targets:

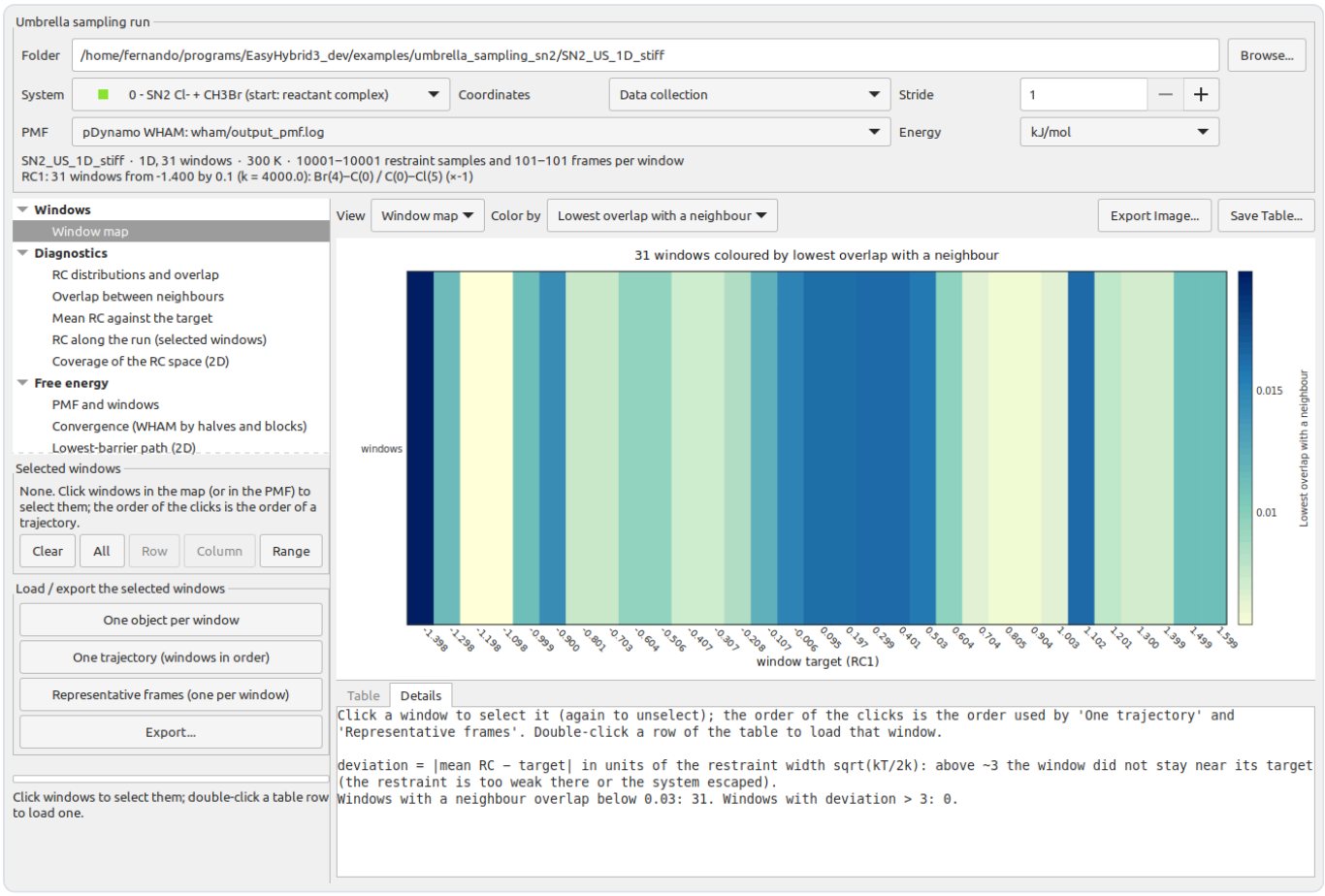


RC distributions of the stiff run: narrow, separated peaks



Overlap of neighbours in the stiff run: all 30 pairs below 0.03

All 30 pairs are below 0.03. WHAM still runs and still gives a PMF (here with a barrier close to the good one), but nothing in that PMF is determined by the data between the windows: the result cannot be trusted. The map coloured by *Lowest overlap with a neighbour* makes the problem visible at a glance:



The window map of the stiff run coloured by the lowest overlap with a neighbour

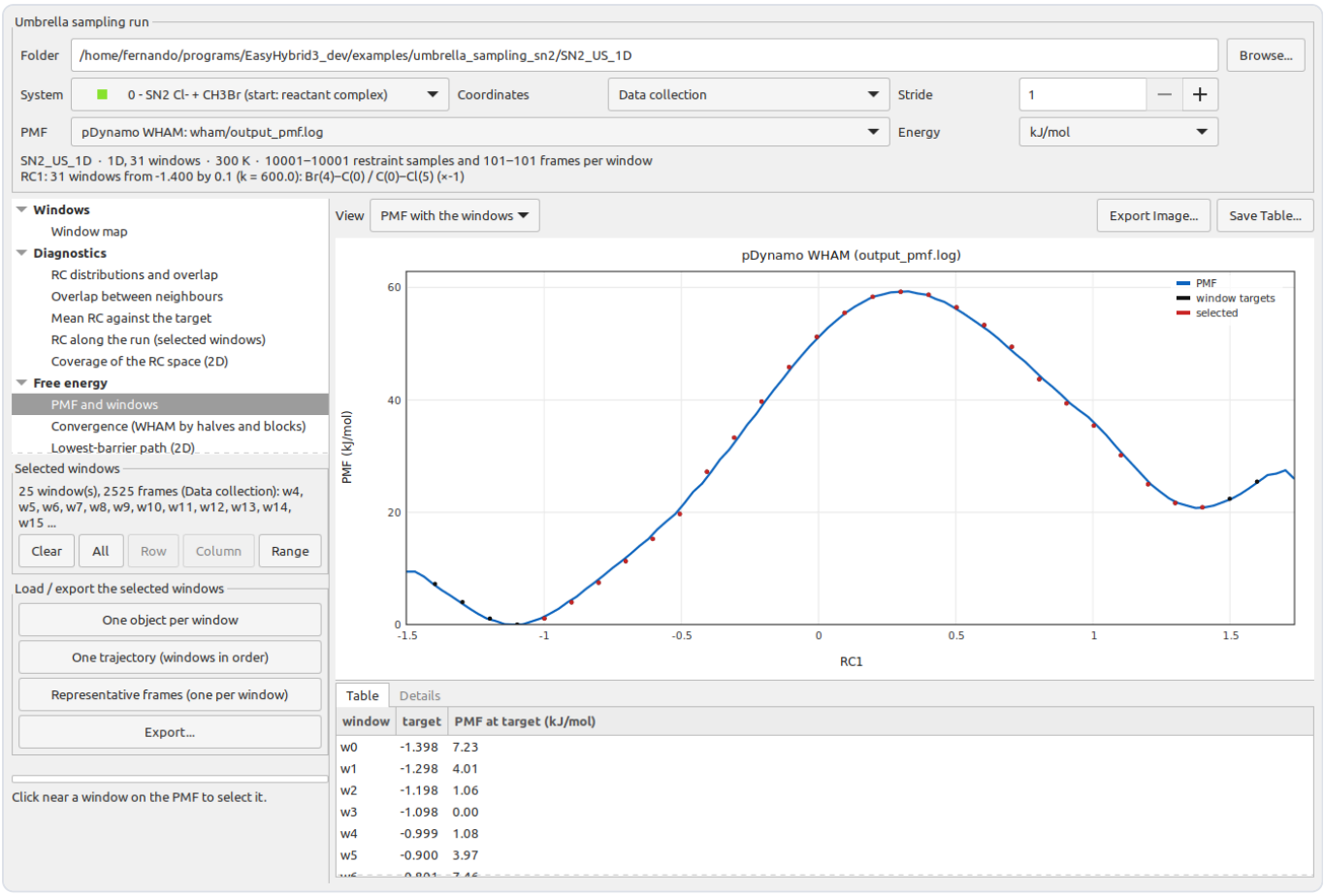
The fix is a softer force constant or more windows. A rule of thumb: the width of a window, sqrt(kT/2k), should be at least about half the distance between targets. At 300 K, kT = 2.49 kJ/mol, so for windows 0.1 Å apart k should not exceed about 500–600 kJ mol^{-1} Å^{-2}.

4. The free energy

Go back to `SN2_US_1D`.

PMF and windows

Free energy → PMF and windows plots the PMF chosen in the *PMF* box (here the file of pDynamo's WHAM, `wham/output_pmf.log`) with the window targets on it. Clicking near a window selects it.

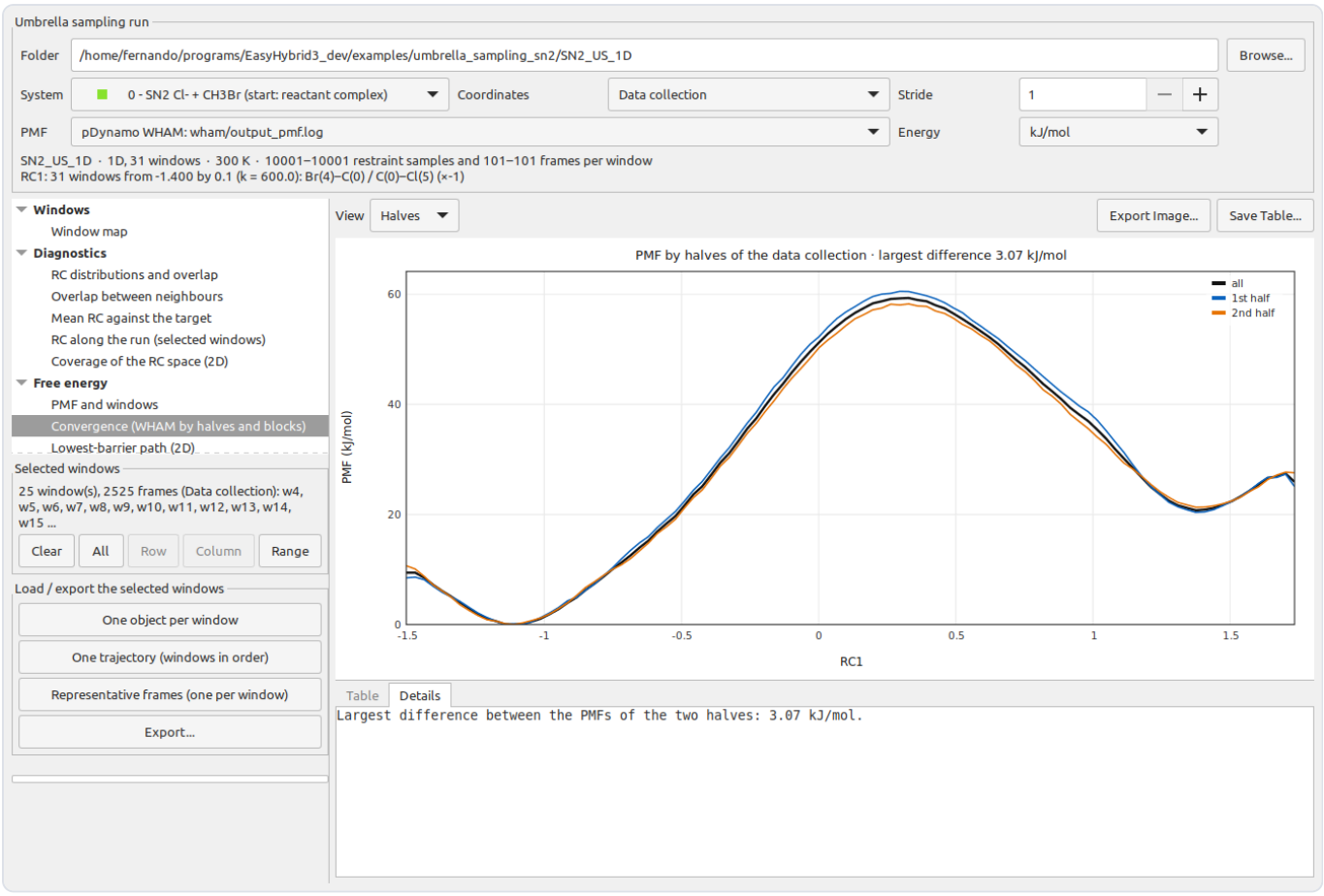


The PMF of the 1D run with the window targets; selected windows in red

The reactant complex is at RC ≈ −1.1 Å, the barrier at RC ≈ +0.3 Å, about 59 kJ/mol higher, and the product complex ($ClCH_3 \cdots Br^-$) at RC ≈ +1.4 Å, about 21 kJ/mol above the reactant complex.

Convergence

Free energy → *Convergence* computes the PMF again from the first half and from the second half of every window (and, in 1D, from each quarter), with a WHAM implemented in the browser, and compares them. Press **Compute** (a few seconds in 1D, about 20 s per PMF in 2D).



PMF of the 1D run from the first and the second half of the data

The two halves differ by at most 3.1 kJ/mol, mostly near the barrier: the run is reasonably converged. A large difference, or a systematic shift between the halves, means the windows need longer data collection or longer equilibration.

> The browser's own WHAM is used to compare halves and blocks with the same bins and equations. The PMF to report is the one of *Analysis → WHAM* (pDynamo), which can also give bootstrap errors.

5. Loading structures

Selecting windows

Click windows in the map (or in the PMF) to select them; click again to unselect. The order of the clicks is the order of the windows in a trajectory. The buttons under the selection:

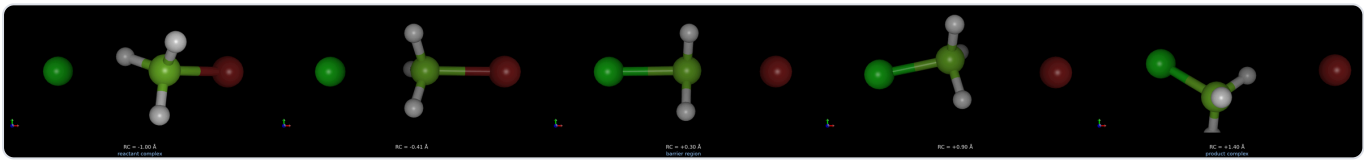
Button	Selects
Clear / All	nothing / every window in order
Range (1D)	every window between the first and the last selected
Row, Column (2D)	the windows with the same RC1 (row) or RC2 (column) index as the last selected

For the figures below, windows 4 to 28 were selected: click window 4, then window 28, then *Range*.

Loading

Button	Result
One object per window	each selected window as an object, with all its frames (source and stride above)
One trajectory (windows in order)	all the frames of the selected windows in one object
Representative frames (one per window)	one frame per window, the one whose RC is closest to the window's target: a "movie" of the reaction
Export...	the representative frames or all the frames to DCD, XYZ or PDB

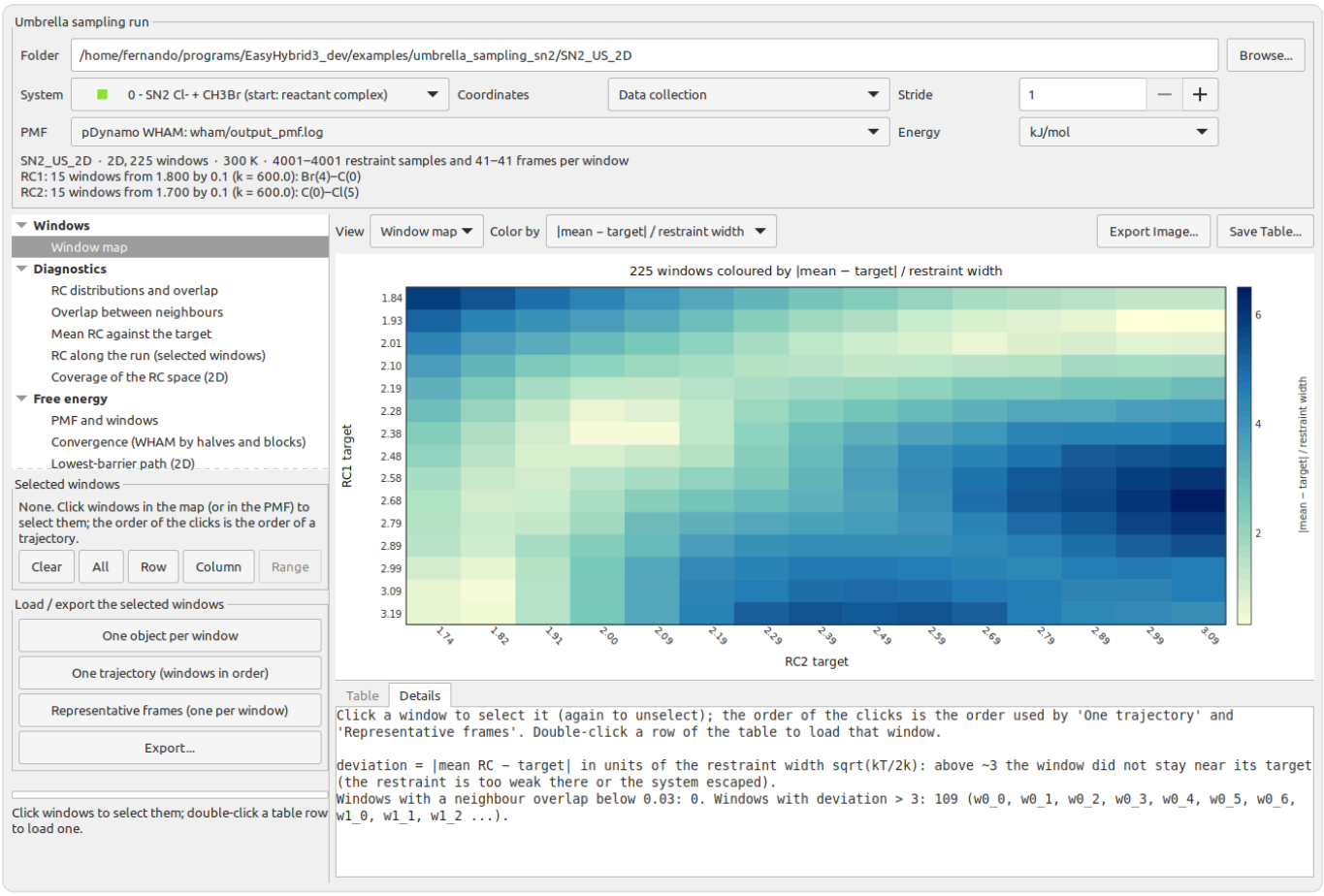
Double-clicking a window in a table loads that window alone. *Representative frames* with windows 4–28 gives an object of 25 frames; the mini player plays it, and the C–Cl bond forms while the C–Br bond breaks and the methyl group inverts:



Representative frames along the 1D path, from the reactant complex to the product complex (spheres and dynamic bonds)

6. A 2D run

Open `SN2_US_2D`. The window map is now a 15 × 15 matrix, rows ordered by the RC1 target (d(C–Br)) and columns by the RC2 target (d(C–Cl)), in the same orientation as the 2D PMF.

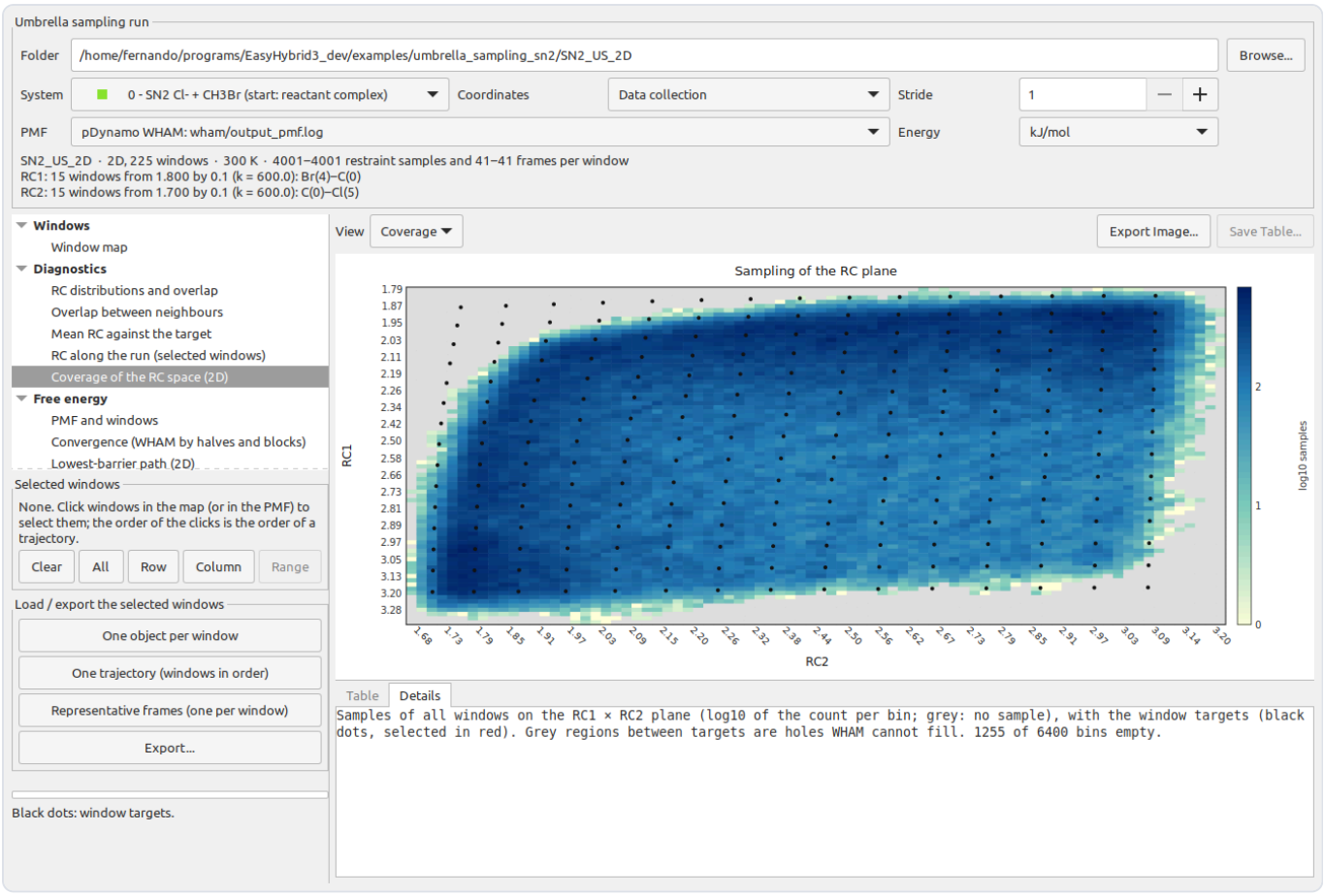


The window map of the 2D run, coloured by the deviation from the target

Two corners are dark: the windows with both distances short (Cl and Br both bonded to carbon) or both long (a free CH_3 between two ions). Those are hundreds of kJ/mol above the reaction path and a force constant of 600 cannot hold them on target; they slide towards lower energy. That does not matter for the reaction, which runs along the other diagonal, but the map shows where the grid was not sampled as planned.

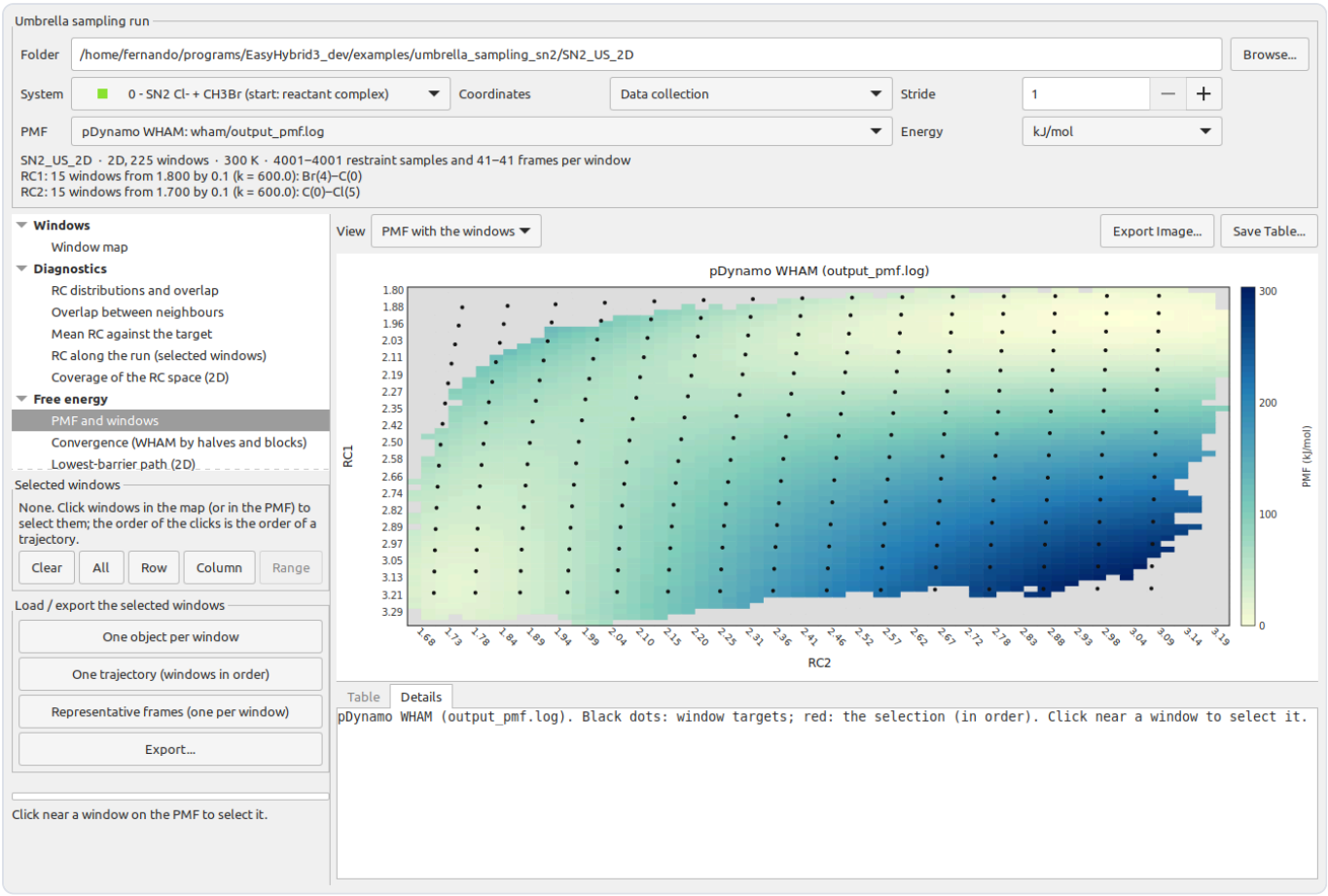
Coverage

Diagnostics → *Coverage* shows all the samples of all windows on the RC1 × RC2 plane (log of the count per bin), with the window targets as black dots. Grey bins have no sample. Holes between targets are regions WHAM cannot fill; here the sampled region covers the whole reaction valley.



Coverage of the RC plane by the samples of all windows, with the window targets

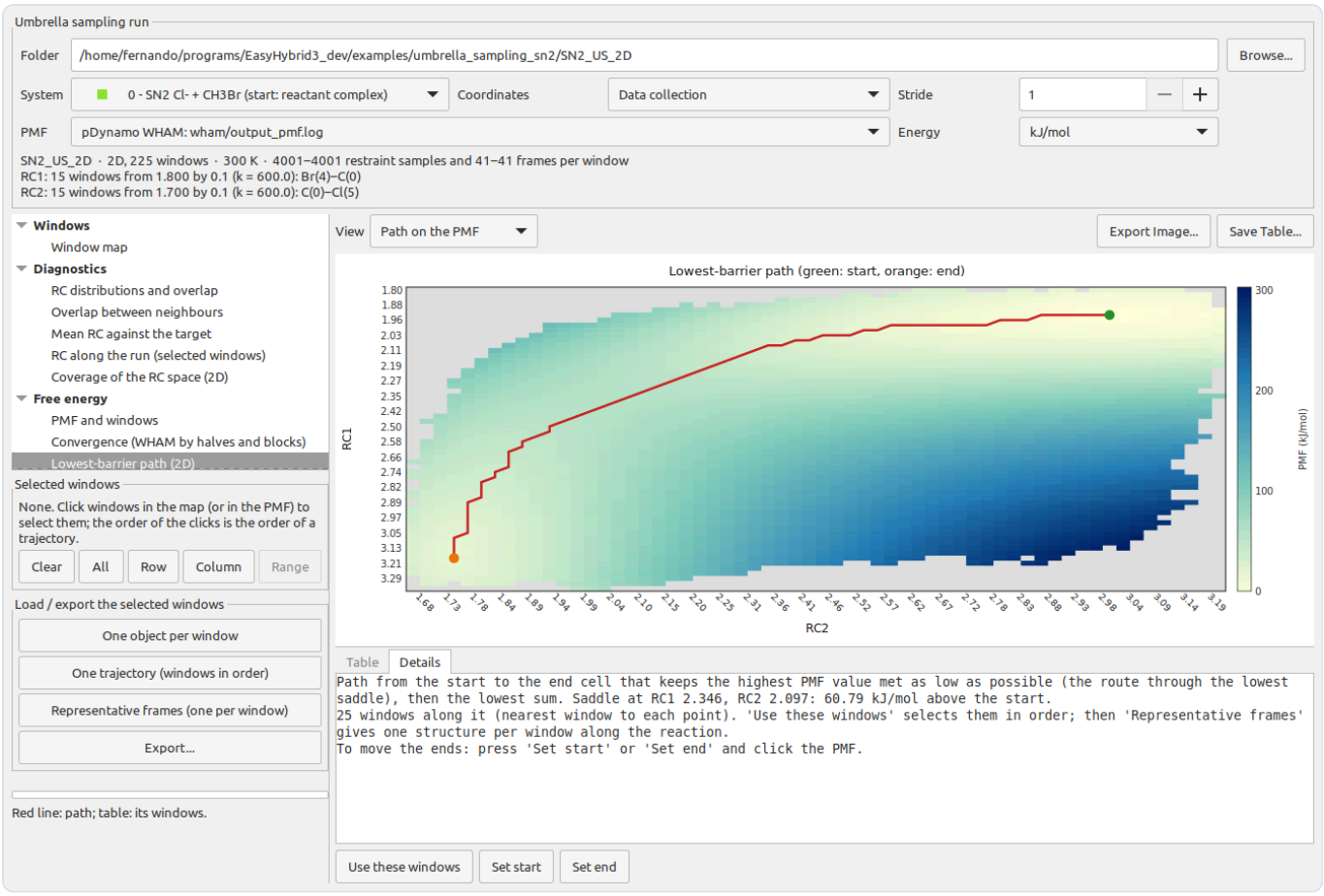
The 2D PMF



The 2D PMF from pDynamo's WHAM with the window targets

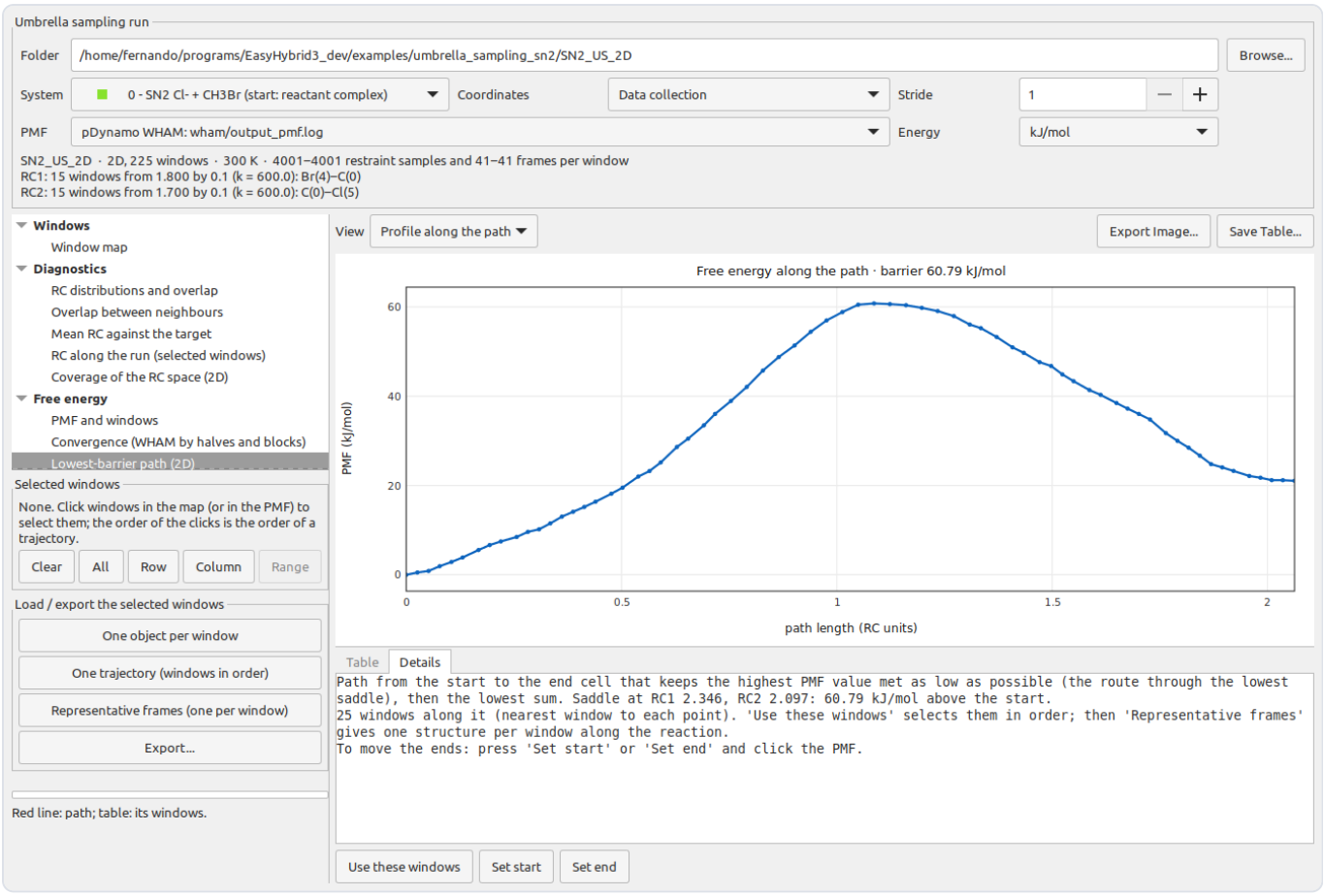
The lowest-barrier path

Free energy → *Lowest-barrier path* finds, on the 2D PMF, the path between two points that keeps the highest free energy met as low as possible: the route through the lowest saddle. By default it goes from the global minimum to the lowest point far from it; *Set start* and *Set end*, then a click on the PMF, move the ends.



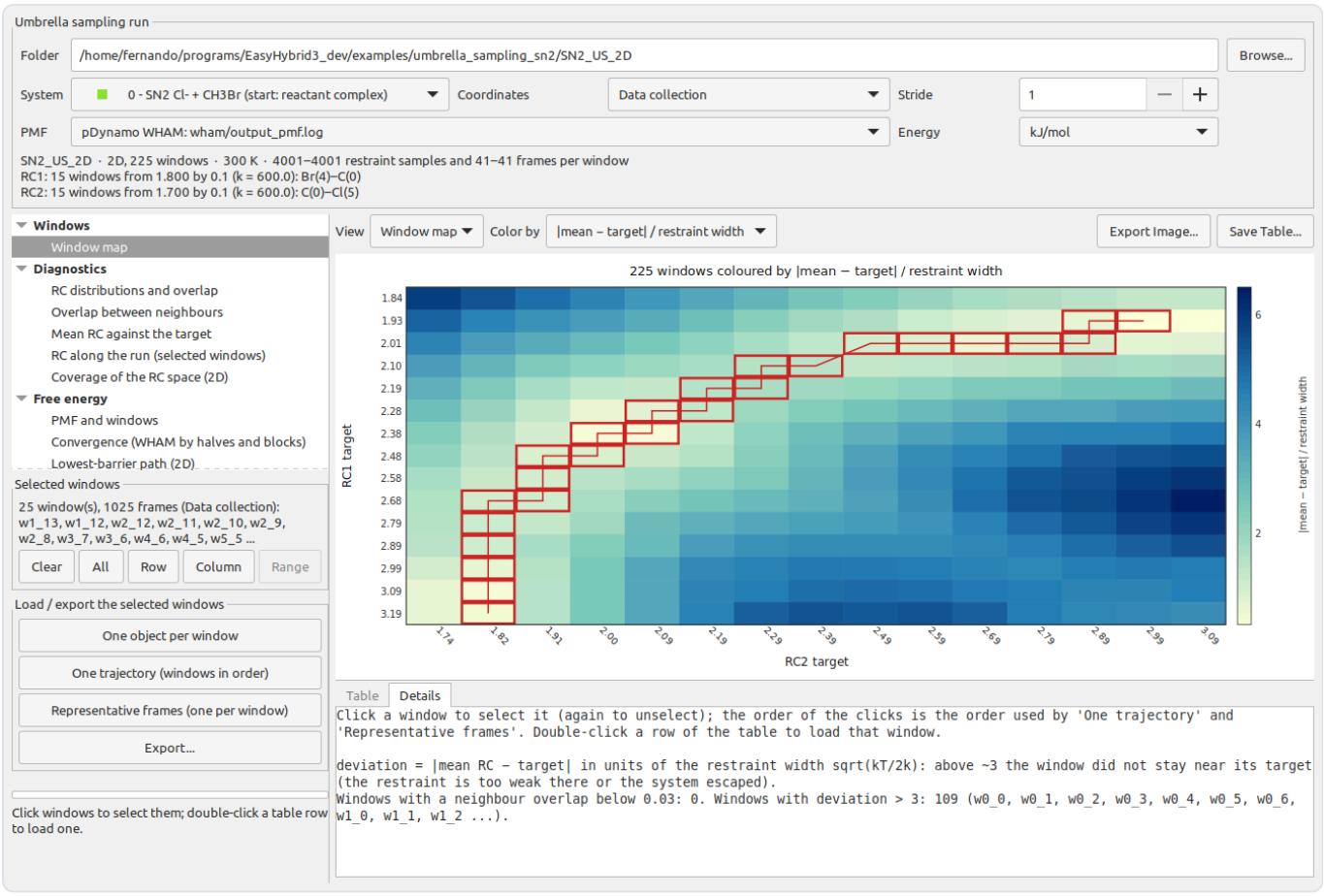
The lowest-barrier path on the 2D PMF, from the reactant complex (green) to the product complex (orange)

The *Profile along the path* view gives the free energy along it: the saddle is at d(C–Br) = 2.35 Å and d(C–Cl) = 2.10 Å, 60.8 kJ/mol above the reactant complex, in agreement with the 1D run (59 kJ/mol).



Free energy along the lowest-barrier path

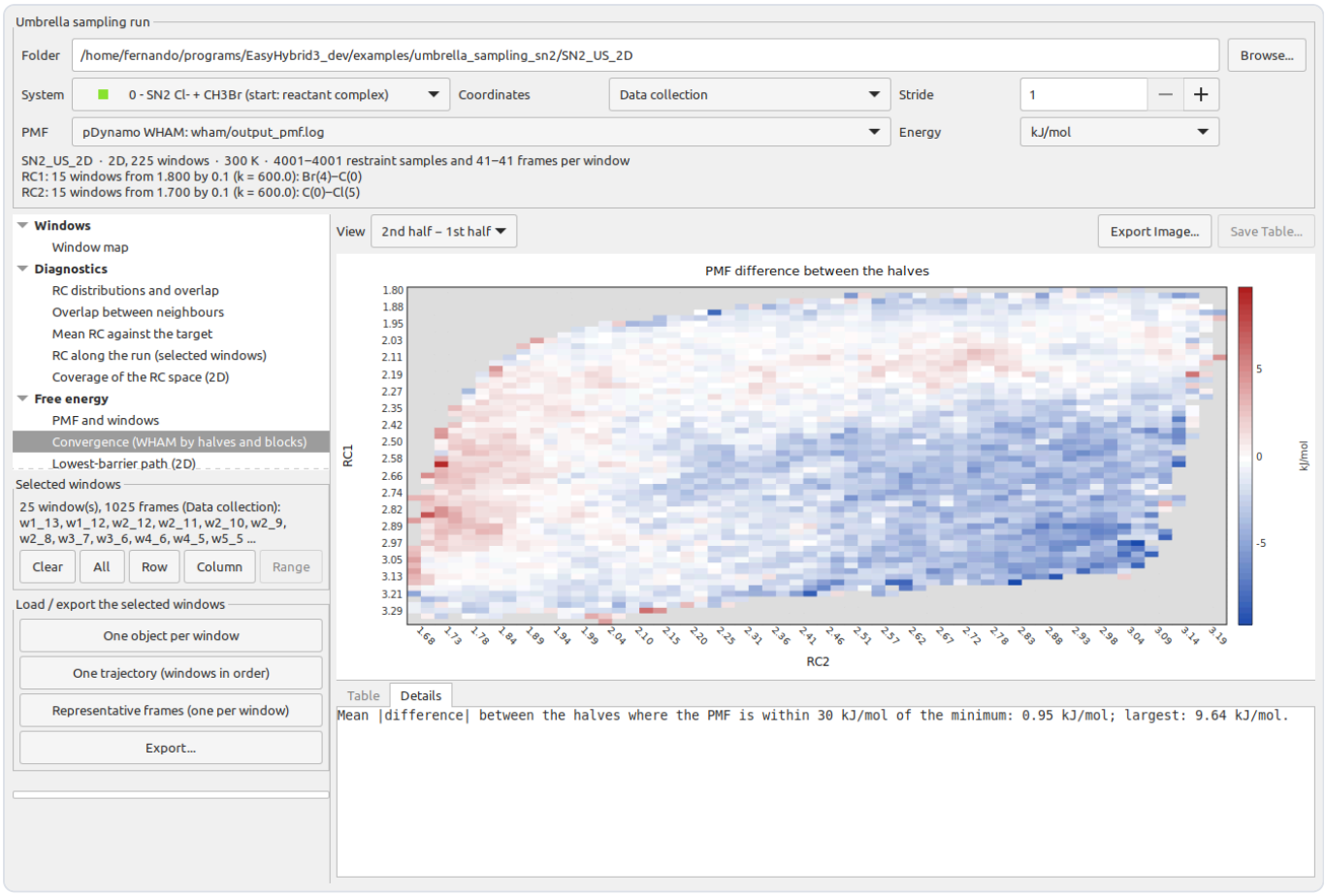
The table lists the windows nearest to the path, in order (25 here). **Use these windows** selects them; the map shows them, and *Representative frames* turns them into a movie of the reaction along the 2D surface.



The windows of the path selected in the map

Convergence in 2D

In 2D the convergence view shows the difference between the PMFs of the two halves. Where the PMF is within 30 kJ/mol of the minimum, they differ by 0.95 kJ/mol on average; the larger differences are at the high-energy edges, which are sampled by few points.



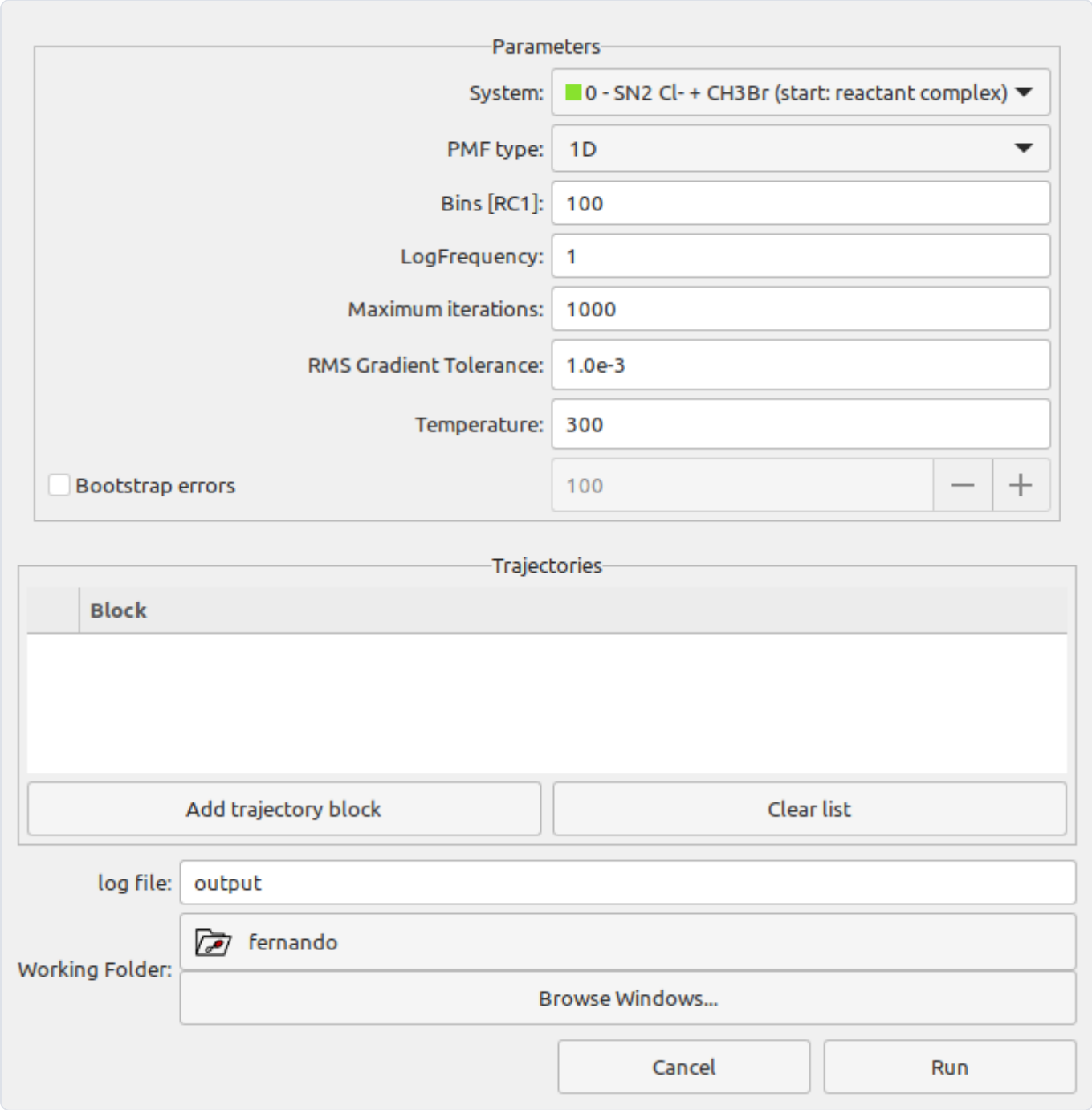
Difference between the 2D PMFs of the second and the first half of the data

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7. Quick reference

Symptom in the browser	Probable cause	What to do
Overlap below 0.03 between neighbours	force constant too large for the window spacing	lower k, or add windows in the gap
Windows with deviation above 3	force constant too small for the slope of the PMF, or the system escaped	higher k there, or accept it in regions off the path
Large drift	equilibration too short	longer equilibration
Halves differ near the barrier	data collection too short	longer data collection
Grey holes in the 2D coverage	windows missing or not sampled	add windows, or restrict the analysis to the sampled region
A window with fewer frames	the window failed	check the output of that window and run it again

Every plot can be saved with *Export Image...* (PNG or SVG) and every table with *Save Table...* (CSV).



The WHAM window: Browse Windows... opens the browser on the chosen run

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EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3